How much can a causal-discovery benchmark resolve? Sampling noise, reporting choices and the simulation gap on real data

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Abstract

Methods for causal discovery are often compared by gains of a few accuracy points on the Tübingen cause–effect pairs or a few edges on the Sachs protein network. We ask what these benchmarks can resolve. On the 102 univariate Tübingen pairs, twelve direction rules have accuracies from 40% to 65%, and only one of 66 paired comparisons survives a Holm correction; a paired test at $n = 102$ has 80% power only for differences of about 19 points (28 after a Bonferroni correction over 66 tests). Across the ten defensible reporting choices that leave at least 30 pairs (official weights or none; subsets by sample size; evaluation on the most confident 25 to 100% of pairs) the accuracy of a single method moves by 9 to 26 points and its rank by up to ten places, and four of twelve methods are ranked first under at least one choice. Trying ten random subsets of 50 pairs raises the expected best accuracy by seven to eight points (a finite-sample selection effect that does not change the ranking). On the Sachs network, 16 of 22 methods have a bootstrap interval that contains the best F_1, and the smallest significant difference is 0.17. Methods trained on simulated data lose the most: a pair classifier falls from 95% on its simulator to 40% on Tübingen. We identify four sources that can produce apparent gains without any change in a method's accuracy, none of which requires a fault of the original authors (we audited no published table), and give a reporting checklist: paired tests with a power statement, the full coverage curve, subsets fixed before the data are seen, and a statement of the effective sample size. We re-implemented four methods from 2025–2026 preprints and ran nine published packages at their defaults; none is distinguishable from the best simple baseline at this power (an absence of evidence, not a demonstration of equivalence).

1 Introduction

New causal-discovery methods are usually accompanied by a table in which the new method is best. On pairs the table is the Tübingen collection of 108 cause–effect pairs [Mooij et al., 2016], 102 of which are univariate; on graphs it is the Sachs protein network [Sachs et al., 2005] with 17 consensus arcs, or simulated graphs. We do not question any particular table. We ask a prior question: with the data these benchmarks contain, which differences between methods can be detected at all, and how far can a correct but unconstrained choice of protocol move a reported number?

The question is not academic. A recent same-hands re-evaluation of bivariate rules on Tübingen with a parameter-free compression baseline reports numbers close to ours for RECI and IGCI [Stienstra, 2026], and benchmark artefacts that favour simple baselines on simulated graphs are documented [Reisach et al., 2021]. We add three things: a quantitative resolution analysis on real data (Section 3), a multiverse over reporting choices that shows how large the room for selection is (Sections 4, 5), and evidence that learned methods lose their advantage when the test data differ from the simulator (Section 6). The methods compared were all run by us under one protocol; they are listed in Section 2.

What this paper does not claim. We did not audit any published table, we do not know which protocol an author used, and a method that is not distinguishable from a baseline here may still be better on other data. The multiverse is a set of choices that a reader of a paper could face; it is not a statement about what authors did.

2 Data and methods

Pairs. The 102 univariate Tübingen pairs with the official weights (the weights downweight groups of similar pairs). Preprocessing is fixed: at most 2000 random rows, each variable dequantised with a small uniform jitter.

Methods. Twelve direction rules, all run by us at settings fixed before the evaluation: IGCI [Daniušis et al., 2010], RECI [Blöbaum et al., 2018], an additive noise model with a spline regression and an HSIC test, the pairwise LiNGAM statistic [Hyvärinen and Smith, 2013], a Gaussian-process additive noise model, LOCI [Immer et al., 2023] with the authors' code, a classifier trained on simulated pairs (PDM, ours), sorted-conditional compression [Stienstra, 2026] on dequantised and on raw rows, topological residual asymmetry [El Bouchattaoui, 2026], AVICI [Lorch et al., 2022] with its pretrained model on two-variable input, and RESIT with a random forest. Compression, TRA and AVICI/RESIT are recent; compression and TRA were implemented from the preprints because no code was available (TRA's constants are our choice).

Graphs. The Sachs observational data (853 cells, 11 proteins, 17-arc consensus network) and the Causal Chambers [Gamella et al., 2025] wind- and light-tunnel recordings with a known graph. Twenty-two graph methods or configurations are on Sachs (PC, GES, DirectLiNGAM, NOTEARS, DAGMA, PCMCI+, GOLEM, NOTEARS-MLP, GraN-DAG, DAG-GNN, GRaSP, BOSS, RESIT, CAM-UV, AVICI, BF-BIC with BOSS [Ramsey et al., 2025], and our own PC-GCM, BTAD, LRG and sequential certificates). The full protocol and all per-method tables are in the companion evaluation (`paper/bench` in the repository).

Protocol. Each evaluation script was committed before it was run (experiments 42–48 in the repository). The multiverse and subset analyses (experiment 47) were designed after the per-pair decisions existed, and were revised once after an independent review; the first version is in the repository history.

3 Sampling noise

Table 1 gives accuracies with exact Clopper–Pearson intervals; the half-width is about 0.10 for every method. Two methods that agree on about 56% of the pairs can differ in accuracy, and the exact McNemar test on the discordant pairs is the correct comparison because the methods are scored on the same pairs. Of 66 pairwise comparisons, 11 have a raw $p < 0.05$ and one survives Holm ($p = 0.041$). Simulating a paired comparison with the observed discordance rate (44%), a difference of 0.19 in accuracy is needed for 80% power. A reported gain of two to five points on Tübingen is therefore far below what the collection can resolve, whatever the method.

Table 1: Twelve pair rules on the 102 Tübingen pairs (unweighted accuracy, 95% interval). Multiverse: the range of accuracy over 32 reporting universes, the best and worst rank, the number of universes in which the method ranks first (ties share the rank), the number in which it is above RECI, and the expected best accuracy after trying ten random subsets of 50 pairs.

Method	Accuracy [interval]	Multiverse range	Rank	First	>RECI	Best of 10 subsets
IGCI	0.559 [0.46, 0.66]	0.47–0.71	3–11	0	0	0.638
RECI	0.608 [0.51, 0.70]	0.61–0.87	1–5	6	0	0.683
ANM-HSIC	0.608 [0.51, 0.70]	0.57–0.70	2–12	0	1	0.683
Pairwise LiNGAM	0.608 [0.51, 0.70]	0.54–0.74	3–12	0	0	0.685
PDM	0.402 [0.31, 0.50]	0.40–0.61	10–12	0	0	0.478
LOCI	0.647 [0.55, 0.74]	0.63–0.77	1–9	2	2	0.722
ANM, Gaussian process	0.554 [0.45, 0.65]	0.55–0.75	2–11	0	0	0.627
Compression, dequantised	0.608 [0.51, 0.70]	0.61–0.79	2–7	0	0	0.682
Compression, raw rows	0.647 [0.55, 0.74]	0.64–0.76	1–7	1	2	0.721
TRA (82)	0.646 [0.53, 0.75]	0.65–0.76	1–5	2	4	0.729
AVICI	0.588 [0.49, 0.68]	0.54–0.65	5–11	0	0	0.664
RESIT	0.613 [0.52, 0.71]	0.61–0.70	3–9	0	1	0.685

4 A multiverse of reporting choices

We evaluate every method in the 32 combinations of (i) official weights or none, (ii) all pairs or only pairs with at least 250, 500 or 1000 rows, and (iii) accuracy on all pairs or on the 75, 50 or 25% most confident decisions (by the absolute score of the method). Each is a choice that has been made in the literature on pairs for a reason: weights are the benchmark's recommendation, a row threshold avoids tiny pairs, and decided-only accuracy reflects a method that abstains. The coverage is taken exactly (the top k by |score|, ties broken at random with a fixed seed); a score is not a calibrated confidence for every rule, so this is the operation a reader could apply, not a measure of each method's own abstention. A universe is kept only if every method is evaluated on at least 30 pairs: 10 of the 32 qualify, and the other 22 (for example 25% of the 36 pairs with at least 1000 rows, which is 9 pairs) are too small to report. In a first run that included them the ranges were far wider (0.27 to 0.75 for IGCI, up to 1.00 for RECI); we exclude them because an accuracy on nine pairs is not a defensible report. In the 10 remaining universes a method's accuracy still moves by 9 to 26 points (0.47 to 0.71 for IGCI, 0.61 to 0.87 for RECI, 0.63 to 0.77 for LOCI) and its rank by up to ten places (ANM-HSIC from second to twelfth, pairwise LiNGAM from third to twelfth). RECI is first in 6 universes, LOCI and TRA in 2 each, compression on raw rows in 1; the ranking among the top methods changes with the protocol. The first counts are over universes, not within one universe, and ties share the first rank. TRA returned no value on 20 pairs; these are excluded from its accuracy (it is evaluated on 82) and from its coverage ranking.

Two single choices are large in themselves. The official weights move RECI from 60.8% to 70.9% and compression from 60.8% to 68.0%, while for LOCI they move it down (64.7% $\rightarrow$ 63.9%). Raw against dequantised rows moves compression from 64.7% to 60.8% unweighted.

5 Selective subsets

A common practice in pair benchmarks is to evaluate a method on a subset of pairs (for example those with a given variable type or sample size). We drew 10000 random subsets of 50 pairs. The accuracy of a method on a subset has a standard deviation of about 0.05 and for the eight methods

whose full-sample accuracy is within 0.04 of RECI the probability of beating RECI on a subset of 50 lies between 0.35 and 0.72. The expected best accuracy over m independently drawn subsets, which is what is reported by a researcher who tries m subsets and keeps the best, grows with m: for RECI 0.608 ($m = 1$), 0.649 (3), 0.683 (10); for LOCI 0.647, 0.688, 0.722. The inflation at $m = 10$ is seven to eight points for every method, so a ranking is not changed by it; what changes is the claim of an absolute level. Because each subset is half of the same 102 pairs, this measures selection noise about a known full-sample accuracy, not how a researcher actually chooses subsets (by pair type or size, not at random).

6 The simulation gap

Methods trained on simulations lose heavily, but dataset difficulty is a confound. Our own pair classifier (PDM) has 95% accuracy on its simulated test and 83% on a different simulated family, and 40.2% on Tübingen ($p = 0.059$ against chance; 0.93 and 0.84 on the two chamber pair sets). Our graph classifier (LRG) has $F_1 = 0.86$ on simulated graphs and 0.36–0.53 on the chambers. The pretrained AVICI model reaches 90% on pairs from independent-row chamber recordings but 58.8% on Tübingen and 55.7% on pairs from random-walk recordings. The simple rules also drop between these sets (RECI 98% $\rightarrow$ 61%, IGCI 94% $\rightarrow$ 56%): chamber pairs are easy, and we have no baseline on the simulated test sets, so the comparison is suggestive. PDM is below chance on Tübingen, which we did not diagnose. We cannot say which feature of the real data is responsible; the simulators contain serial dependence, drift and several mechanism families and still do not cover them.

7 Graph benchmarks

On Sachs the same analysis gives the same picture. For 22 methods, 16 have a 95% bootstrap interval (over the 55 variable pairs) that contains the best F_1 (0.67) and 7 have a point estimate within 0.10 of it. Of 231 pairs of methods (some are the same method at two settings), 62 differ significantly and the smallest significant difference in F_1 is 0.17. The reference has 17 arcs, so one edge changes recall by 0.06. In the Causal Chambers, the bipartite graph (actuators to sensors) makes the skeleton mostly feature selection, and in 17 of 19 new wind-tunnel recordings the actuators are random walks with an effective sample size of about three out of 4000 rows, where no method has strict F_1 above 0.42. Several neural methods time out at sixteen variables. Appendix tables of the companion evaluation give the per-method values.

8 Why published gains can look larger

Combining the evidence, a published gain on a small real benchmark can arise without any change in the method's true accuracy from (i) the resolution limit: two methods that differ by two to five points are statistically the same on 102 pairs; (ii) free reporting choices that move one method by up to 26 points even after discarding universes with fewer than 30 pairs; (iii) selection among subsets or settings, worth seven to eight points after ten attempts; and (iv) the gap between a simulator and real data for learned methods. None of these requires an error by an author; each is a degree of freedom the benchmark does not constrain. These are mechanisms, not an accounting of a particular paper.

9 Recommendations

(1) Compare methods with a paired exact test on the same pairs, state the power, and do not interpret a difference below the detectable one. (2) Report the full coverage curve, not one decided-only number, and report weighted and unweighted accuracy. (3) Fix subsets, preprocessing and settings before the data are seen, and publish the code that does it. (4) State the effective sample size of every real data set, and do not use a benchmark whose effective size is a few observations to rank methods. (5) For learned methods, report accuracy on real data next to the simulated test, with the simulator's overlap with the real data described. (6) Prefer benchmarks with more independent units: interventional recordings and collections with many more pairs would lower the detectable difference.

10 Limitations

The study uses 102 pairs and one protein network, which is itself the resolution limit it describes. The twelve methods were run by us; some at library defaults, and tuning may change a ranking. The multiverse is finite, its choices are ours, its accuracies are computed on overlapping pairs and are not independent, the scores of several rules are not confidences, and the Tübingen collection has known label noise and groups of similar pairs (the reason for the weights). The detectable difference (0.19) is for the observed discordance rate (0.44), averaged over method pairs, and is approximate. The graph bootstrap resamples variable pairs that share nodes. We did not run CAM, SCORE or PNL (PNL needs 880 s per pair), and two preprints were implemented from their description. The simulation-gap evidence uses our own learned models and one pretrained model.

Data and code. All code, per-pair decisions and scripts are in the repository `vinhqdang/SPRINT-CD`. The Tübingen pairs, the Sachs data and the Causal Chambers data are public.

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